ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

TxID	147302
ToLID	**aRanIta1**
Species	Rana italica
Class	Amphibia
Order	Anura

Genome Traits	Expected	Observed
Haploid size (bp)	5,101,732,283	5,152,309,785
Haploid Number	13 (source: direct)	13
Ploidy	3 (source: ancestor)	2
Sample Sex	unknown	unknown

EBP metrics summary and curation notes

Obtained EBP quality metric for hap1: 6.8.Q64
Obtained EBP quality metric for hap2: 6.8.Q64

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

. Observed Ploidy is different from Expected

. BUSCO single copy value is less than 90% for hap2

Curator notes

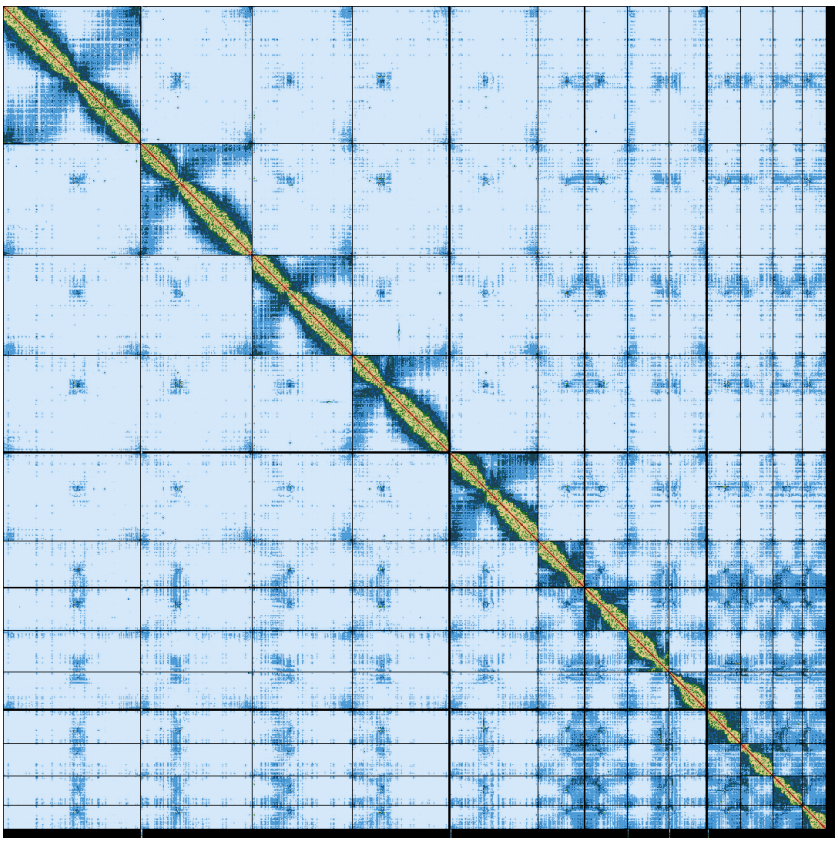
. Interventions/Gb: 36
. Contamination notes: "Both FCS-GX and Blobtoolkit did not detected contaminants in the curated assemblies."
. Other observations: "First, Hifiasm assemblies were scaffolded with yahs without allowing for contig breaks. Then it was curated using the diploid map to produce the curated haplotypes and then reviewed individually with high resolution (HR). In total curation and reviews, made 27 cuts in contigs, 137 breaks at gaps and 206 joins. The links provided point to a shared folder with the High Resolution maps (HR), for each hap or both (diploid), allowing multimappings (mq0) or not (mq10). The savestate_1 for the diploid map has all supers side by side, showing 1:1 correpondence between haplotypes. For hap2, SUPER_5 is named SUPER_5_unloc_1 in the maps, but this little mistake has been fixed in the fasta for the ENA upload. Finally, the mitogenome was assembled into a single circular contig 22,497 bp long that will be uploaded to ENA together with hap1."

Quality metrics table

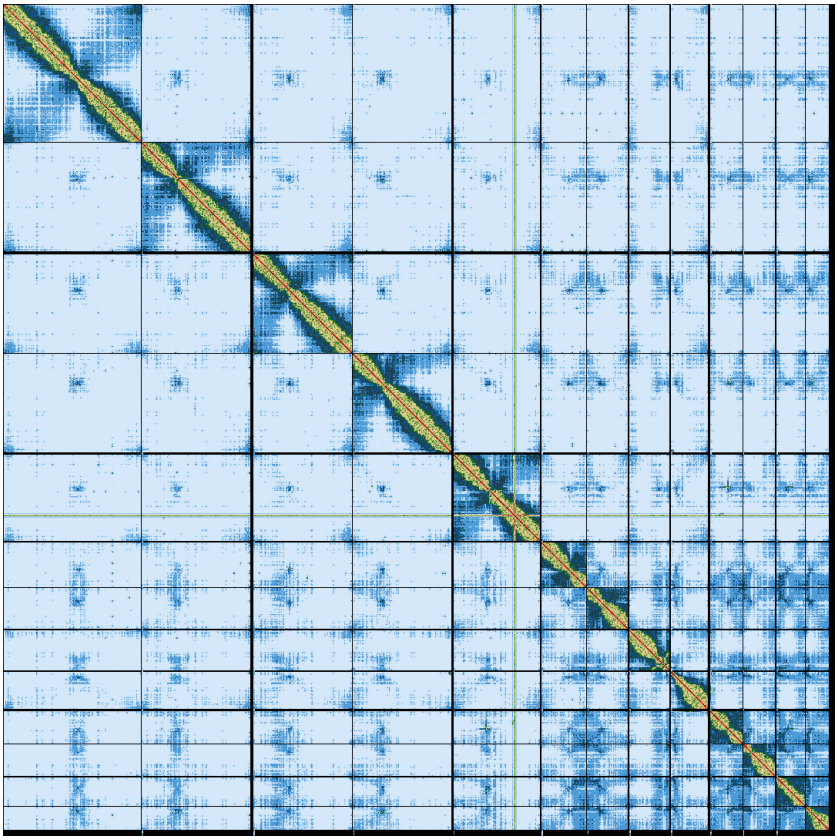
Metrics	Pre-curation hap1	Pre-curation hap2	Curated hap1	Curated hap2
Total bp	5,149,033,152	5,028,367,072	5,152,309,785	5,024,435,688
GC %	44.56	44.5	44.56	44.5
Gaps/Gbp	502.42	532.78	507.15	541.35
Total gap bp	517,400	535,800	522,600	544,000
Scaffolds	344	330	334	297
Scaffold N50	676,956,069	596,098,675	595,068,311	600,065,487
Scaffold L50	3	4	4	4
Scaffold L90	11	11	11	11
Contigs	2,931	3,009	2,947	3,017
Contig N50	5,393,737	4,359,970	5,334,790	4,357,435
Contig L50	279	317	283	319
Contig L90	1,186	1,351	1,200	1,357
QV	64.094	64.3927	64.095	64.3873
Kmer compl.	94.8376	94.0275	94.8789	94.0009
BUSCO sing.	90.1%	87.5%	90.0%	87.5%
BUSCO dupl.	2.9%	2.7%	2.9%	2.7%
BUSCO frag.	1.9%	2.3%	1.9%	2.3%
BUSCO miss.	5.1%	7.5%	5.1%	7.6%

BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: vertebrata_odb12 (genomes:119, BUSCOs:3390)

HiC contact map of curated assembly

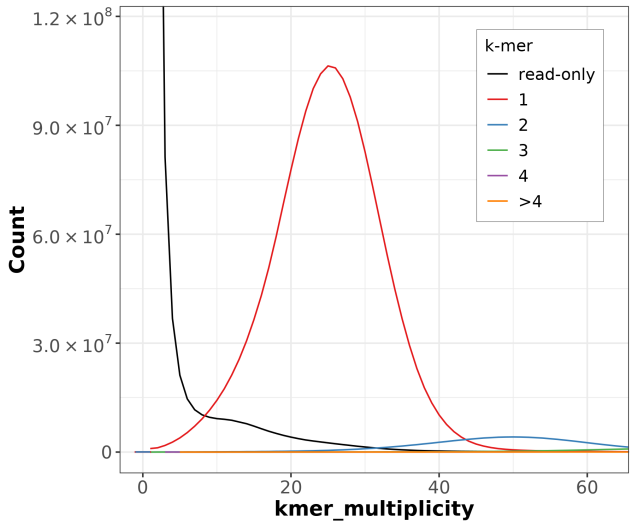


hap1 [LINK]

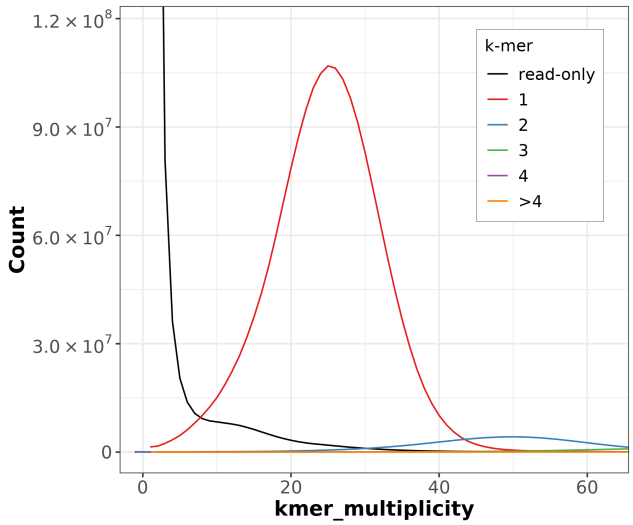


hap2 [LINK]

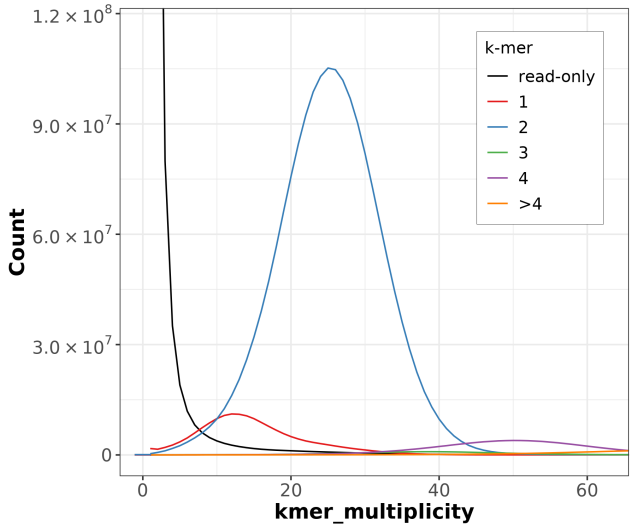
K-mer spectra of curated assembly



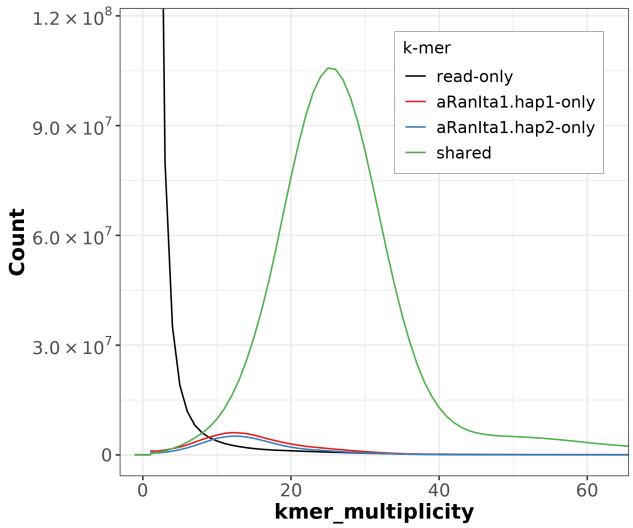
Distribution of k-mer counts per copy numbers found in **hap2** (hapl.)



Distribution of k-mer counts per copy numbers found in **hap1** (hapl.)

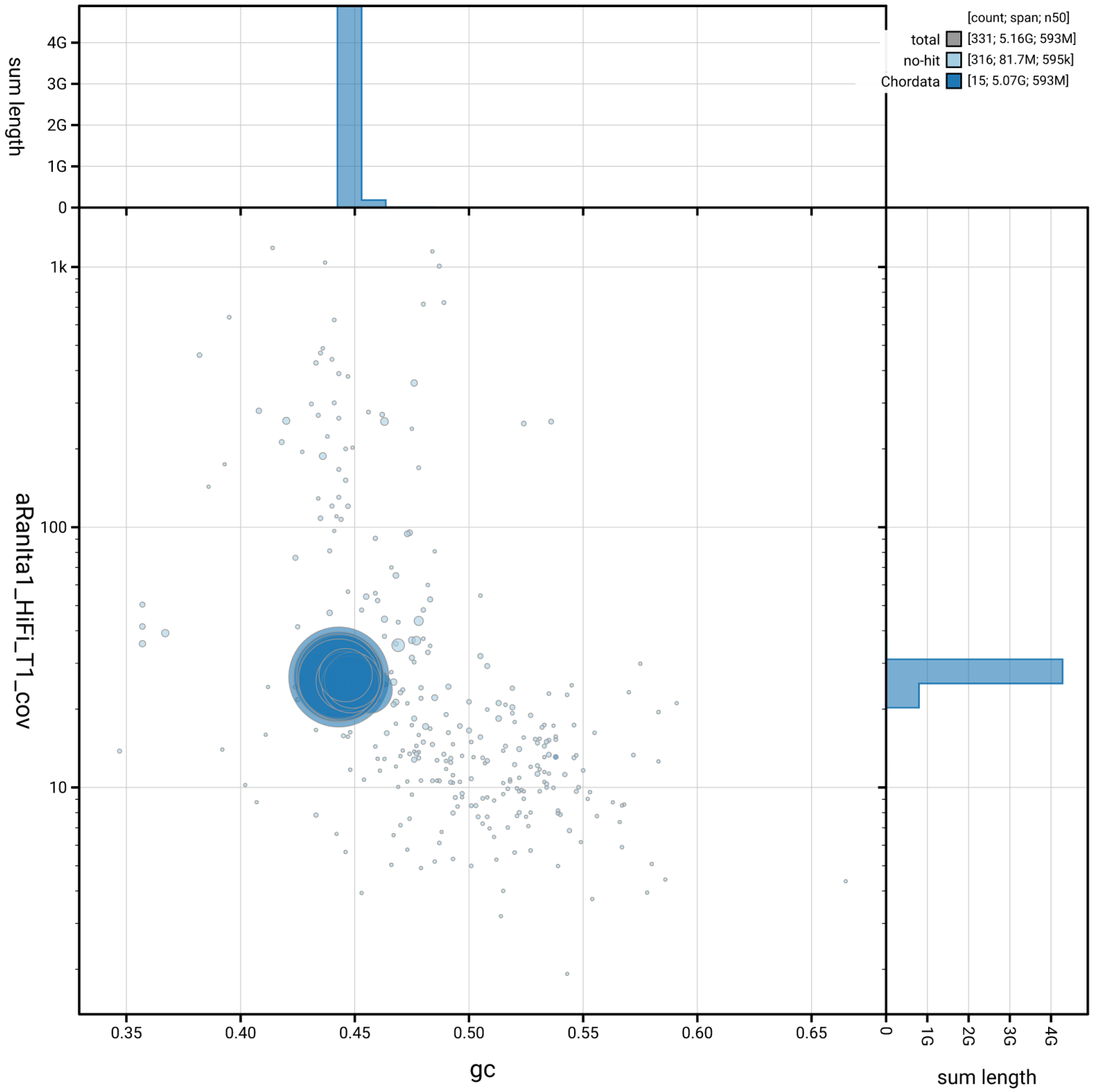


Distribution of k-mer counts per copy numbers found in asm (dipl.)

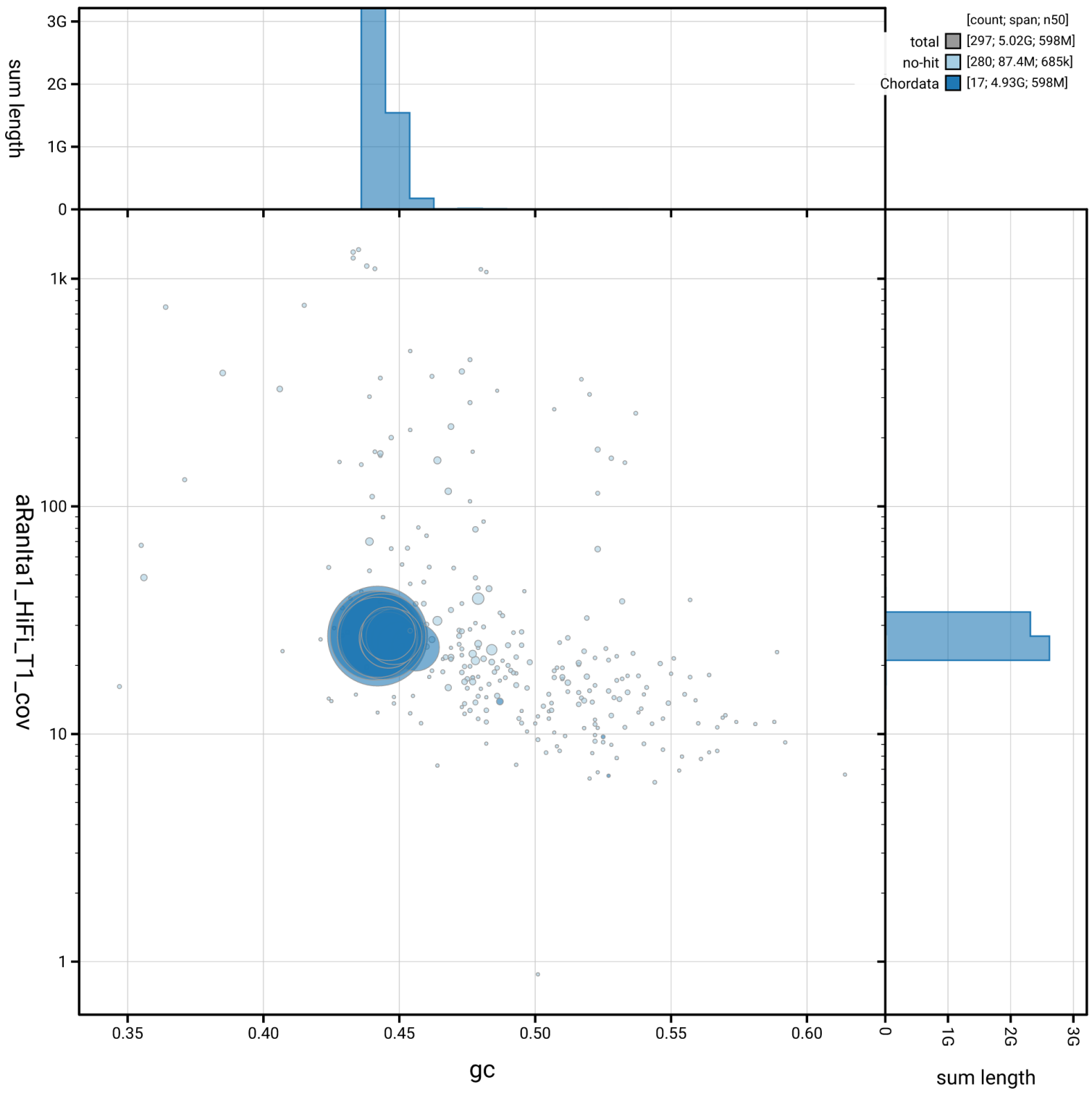


Distribution of k-mer counts coloured by their presence in reads/assemblies

Post-curation contamination screening



hap1. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.



hap2. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	HiFi	Hi-C
Coverage	27x	240x

Assembly pipeline

- **CLAWS pipeline**
    |_ *ver:* 3.1
    |_ *key param:* NA
- **Hifiasm**
    |_ *ver:* 0.24.0
    |_ *key param:* --telo-m TTAGGG
- **YaHS**
    |_ *ver:* 1.2a
    |_ *key param:* --no-contig-ec
- **Merqury**
    |_ *ver:* 1.3
    |_ *key param:* NA
- **BUSCO**
    |_ *ver:* 6.0.0
    |_ *key param:* NA
- **tidk**
    |_ *ver:* 0.2.65
    |_ *key param:* NA
- **chromap**
    |_ *ver:* 0.3.2
    |_ *key param:* NA
- **FOAM pipeline**
    |_ *ver:* 0.5
    |_ *key param:* NA
- **mitos**
    |_ *ver:* 2.1.3
    |_ *key param:* NA

Curation pipeline

- **Blobtoolkit DToL Nextflow pipeline**
    |_ *ver:* 0.6.0
    |_ *key param:* NA
- **FCS-GX**
    |_ *ver:* 0.5.5
    |_ *key param:* NA
- **PretextViewAI**
    |_ *ver:* 1.0.5
    |_ *key param:* NA
- **SANGER tola-agp-tpf-utils**
    |_ *ver:* 1.2.1
    |_ *key param:* NA

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